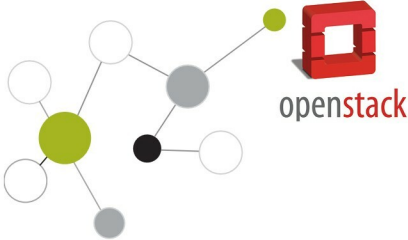


Integrating OpenDaylight VTN Manager with OpenStack



OpenDaylight is the largest open source SDN controller. The OpenDaylight virtual tenant network (VTN) is an application that provides a multi-tenant virtual network on an SDN controller. This article is a tutorial on how to integrate the OpenDaylight VTN Manager with OpenStack.

A virtual tenant network (VTN) allows users to define the network with a look and feel of the conventional L2/L3 network. Once the network is designed on VTN, it automatically gets mapped into the underlying physical network, and is then configured on the individual switch, leveraging the SDN control protocol. The definition of the logical plane makes it possible not only to hide the complexity of the underlying network but also to manage network resources better. This reduces the reconfiguration time of network services and minimises network configuration errors.

The technical introduction given above might seem complex to SDN beginners. In this article, I have tried to be as simple as I can, while teaching readers about VTN and its integration with OpenStack.

For the purpose of this article, I'm assuming readers have a basic understanding of SDN. So let me start with VTN directly.

The SDN VTN Manager helps you to aggregate multiple ports from the many underlying SDN-managed switches (both physical and virtual) to form the single isolated virtual tenant network (VTN). Each tenant network has the capability to function as an individual switch.

For example, consider that you have two physical switches (say, s1 and s2) and one virtual Open vSwitch (say, vs1) in your lab environment. Now, with the help of the VTN Manager, it is possible to group (aggregate) the three ports (say p1, p2 and p3) from switch s1 — i.e., s1p1, s1p2, s1p3; two ports from switch s2 — i.e., s2p1 and s2p2; and two ports from the virtual switch vs1 — i.e., vs1p1 and vs2p2, to form a single switch environment (say, VTN-01).

This means, virtually, the group (tenant) named VTN-01 is one switch with seven ports (s1p1, s1p2, s1p3, s2p1, s2p2, vs1p1 and vs2p2) in it. This VTN-01 will

act exactly like a single isolated switch with the help of flows configured in the ports of all three switches by the OpenDaylight VTN Manager.

The above example explains the concept called port mapping in VTN, and will help beginners to understand the basic concept better. It will also help them to compare all other VTN concepts like VLAN mapping and MAC mapping.

VTN OpenStack integration

There are several ways to integrate OpenDaylight with OpenStack. This article will focus on the method that uses VTN features available on the OpenDaylight controller. During integration, the VTN Manager works as the network service provider for OpenStack.

The features of VTN Manager empower OpenStack to work in a pure OpenFlow environment, in which all the switches in the data plane are an OpenFlow switches. You could also refer to my blog on 'OpenDaylight Integration with OpenStack using OVSDB' from the link

<http://www.cloudenablers.com/blog/opendaylight-integration-with-openstack/>.

The requirements are:

- OpenDaylight Controller
- OpenStack Control Node
- OpenStack Compute Node

OpenDaylight support for OpenStack network types

Till the Boron release, OpenDaylight (ODL) only supported 'Local' network type in OpenStack and there was no support for VLAN. You may wonder why the developers never speak about VxLAN and GRE tunnelling network types support. You can answer that question if you recall the example I mentioned at the beginning of this article.